

CSC621 Data Structures and Algorithm

Lab Exercise 3 - Stacks

Question 1

You are required create a Java/C++ program to implement stacks utilizing arrays in order to change decimal numbers into binary.

Example: converting 28_{10} to Binary would be:

$28 / 2 = 14 \text{ R } 0$	
$14 / 2 = 7 \text{ R } 0$	
$7 / 2 = 3 \text{ R } 1$	
$3 / 2 = 1 \text{ R } 1$	
$1 / 2 = 0 \text{ R } 1$	
The Answer is: 11100	

Question 2

You are required create a Java/C++ program to implement stacks utilizing arrays to change binary numbers into decimal.

Example: converting 11100_2 to Decimal would be:

0	*	2^0	=	0
0	*	2^1	=	0
1	*	2^2	=	4
1	*	2^3	=	8
2	*	2^4	=	<u>16</u>
				28